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Purposefully different,

consistently excellent

Welcome to EEN1037: Web Applications Development

Cascading Style Sheets (CSS)

Week 2

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HTML is not enough

- HTML is intended to display the contents of documents
- Tags for text formatting were not part of HTML initially
- HTML 3.2 specification included tags like <font>
- Adding format information to every single page was tedious

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Cascade Style Sheet

- Separates contents from presentation
- ***Style*** deals with the presentation of web pages e.g. font, color, layout etc.
- Often described in a separate ***sheet*** and saved as a CSS file.
- Multiple styles can be specified and final application of rule is applied by CSS ***cascading*** logic.

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Cascade Style Sheet

- CSS allows much richer document appearance
- Visual appearance is separated from the contents
  - HTML for contents
  - CSS for presentation
- Reduced workload
- Same CSS can be applied on multiple pages
- Reduce page download size

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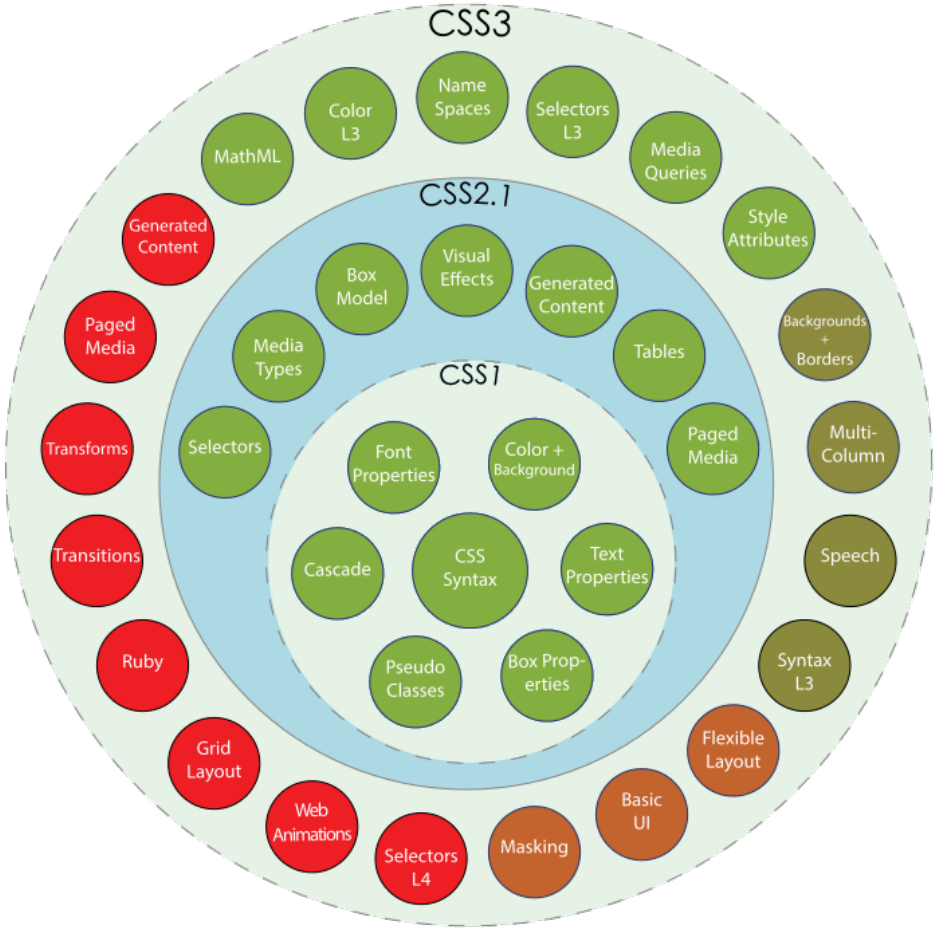
CSS 3

- CSS1 and CSS2 were full specs covering all elements
- CSS3 being released as individual modules

- Recommendation
- Candidate Recommendation
- Last call
- Working draft



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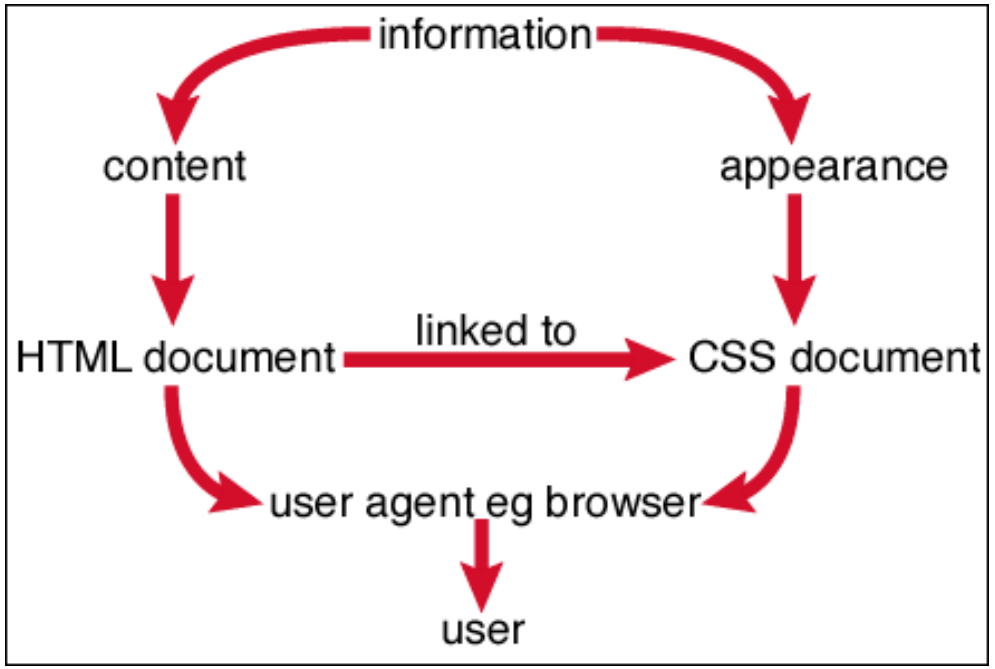
Cascading StyleSheets (CSS)

- **Content from Appearance –** important basic principle



- **Style Benefits –** larger array of layout and presentation options, not in HTML
- **Single Style File –** one file describing layout for entire site

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Cascade Style Sheet

- Linking HTML document to CSS file
  - External style sheet
    - `<head> <link rel="stylesheet" type="text/css" href="myStyleSheet.css"> </head>`
  - Internal style sheet
    - `<head> <style> h1 {color: blue;} </style> </head>`
  - Inline style sheet
    - `<h1 style="color:blue;">`
      `This is a heading in blue color`
      `</h1>`

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CSS Structures

- CSS file handles style and layout, HTML file handles content
- CSS files are made up of rules
- Rules are made up of two parts: selector (e.g. ‘H1’) and declaration (e.g. font-family etc.)
- Declaration split up into a number of name-value pairs (properties)
- Inheritance: `<H1>A Really <B>Ugly Web</B> Page</H1>`
- If we define two style entries in the CSS file, Ugly Web inherits all of the properties of H1, but will override with its own definitions
- To increase granularity of control over elements, we introduce the class attribute. Example. B.BLUE inherits properties from B

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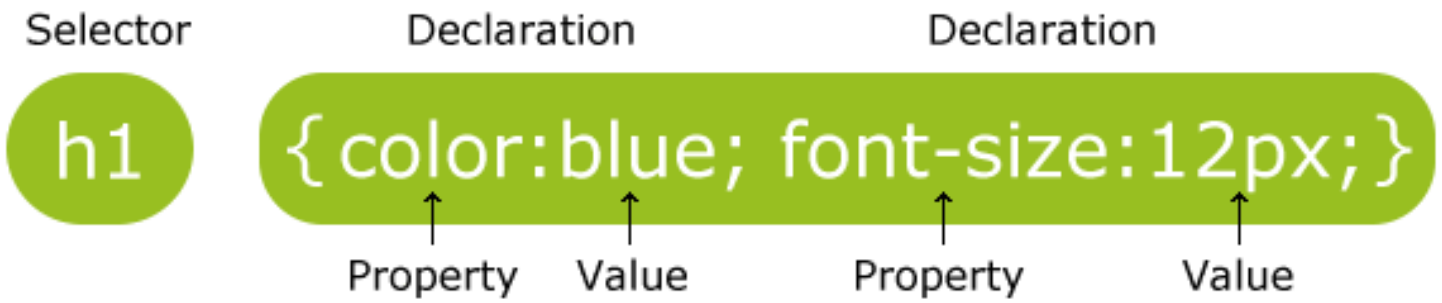
Cascade Style Sheet

- CSS Rule



- *Selector* selects the relevant HTML element to style
- *Declaration* Block contains one or more style declarations containing property name and value

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Cascade Style Sheet

- Selectors
  - Element Selectors: selects all elements of a specific type e.g. `<h1>, <b>`
  - Class Selectors: selects all elements that belong to a given class e.g. `h2.myClass, i.myClass`
  - ID Selectors: selects a single element that have been given a unique id e.g. `<p id="myId">`
  - Pseudo Selectors: combines a selector with a user activated state e.g. :hover, :link, :visited

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Cascade Style Sheet

- Selectors
  - Element Selectors: selects all elements of a specific type e.g. <h1>, <b>

```
h1 {
    text-align: center;
    color: red;
}
```

Applies to all heading <h1> tags in the html document

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Cascade Style Sheet

- Selectors
  - Class Selectors: selects all elements that belong to a given class e.g. h2.myClass, i.myClass

```
p.left {
    text-align: left;
    color: blue;
}
```

Applies to all elements that belong to class left e.g. `<p class="left">` This paragraph will be left aligned `</p>`

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Cascade Style Sheet

- Selectors
  - ID Selectors: selects a single element that have been given a unique id e.g.

```
#myStyleId {
     font-size: 30px;
     font-family: "Times New Roman";
    }
```

Applies to the element with given id e.g. Font size and style for this paragraph only

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Cascade Style Sheet

- Selectors
  - Pseudo Selectors: combines a selector with a user activated state e.g. :hover, :link, :visited

```
a:hover {
    color: red;
    }
```

```
Applies to a link element and changes
color to red when user hovers the mouse
over the link e.g. <a href="page2.html">
Link to my second page </a>
```

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CSS for Layout

- Not only can we make style changes, but we can layout entire sites
- Two main elements to help us with layout in CSS
  - - can apply styles to whole sections of HTML elements

    attributes are *style, class and id*

```
<h2>Div Example</h2>
<p>
Some text before the div we are demonstrating...
<div id="news_section" style="font-size:8pt" class="news">
        Today in the news were a number of startling
revelations...
</div>
Some text after the div we are demonstrating....
</p>
```

  - In a corresponding CSS file:

```
div.news { font-family: Arial, Verdana; color: black;
            background: #eeeeee; margin:20px; padding:10px; }
```

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CSS for Layout

- - works very similarly to but does not apply any formatting of it's own. effectively adds new lines before and after itself

- This means that can be used inline inside elements like paragraphs without affecting the flow of the paragraph.

- See div_span_example.html

Two types of element

- block – these are elements such as or <p>. They start their content on a new line and end on a new line. They can contain other block or inline elements. Eg.
  `<p>Some of <b>this is bold</b></p>`

- inline – these are elements such as ,<i> or <b>. They define style but do not cause any extra formatting or new lines.
  They typically only ever contain other inline elements or text.

  One can set widths and heights for block elements, but not for inline.

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Creating Page Structures

- Traditionally pages/frames or tables were used to create layouts such as columns
- Bad practice
- We can use `<div>` tag for layout. Defaults to 100% width.
- We are able to set widths and heights

  - in units
  - as a percentage of available space

- What should a fixed width be in a webpage? 800, 1000, 1200?
- float property tells the browser to position a block element to the left or right of an existing element (if there is space)
- What happens if we make a column wider, introduce padding, borders?

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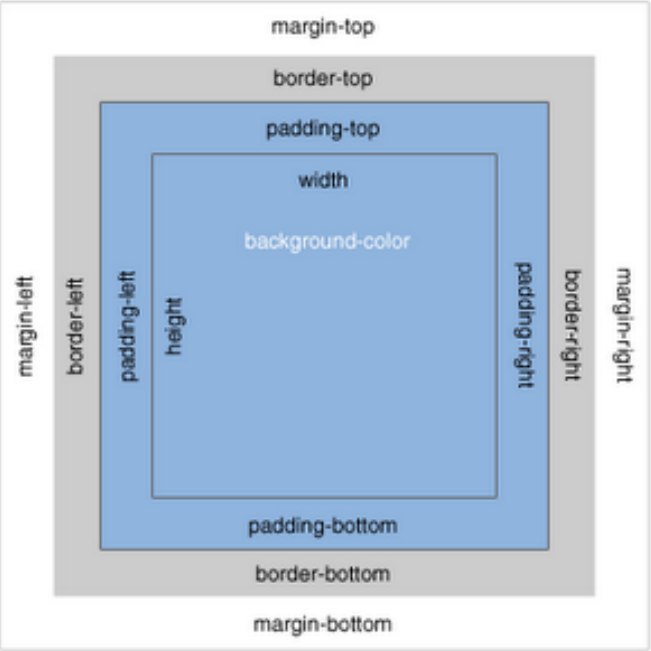
Creating Page Structures

The Box Model

- All HTML ‘block’ elements follow the box model with 5 spacing props:
  - width
  - height
  - margin
  - border
  - padding



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Cascade Style Sheet

- Declaration
  - Properties and Values
- Different properties to format
  - Background: color, image
  - Text: color, align, decoration, transform
  - Fonts: family, style, size
  - List & Tables: list-style-type, table border
- W3Schools tutorial for complete list of properties and values: http://www.w3schools.com/css/default.asp

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Cascade Style Sheet

- Cascading logic if multiple styles are defined for same element
  - All styles will cascade into a new “virtual style” in following order (lower to higher priority)
    1. Browser Default
    2. External Style Sheet
    3. Internal Style Sheet
    4. Inline Style

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